Test Intelligent Systems COMPUTERROOMS

Number of questions: 32
Session period: 24 november 2023 15:30 - 18:30
Duration: —

Instruction

Exam duration: 2 HOURS (120 minutes) unless you get extra time.

The following tools are permitted:
- Calculator
- Scrap paper (nothing written or printed on it)
- Check here for the formula sheet

This first partial exam consists of 32 questions of varying complexity. In total you can reach maximally: 120 Points. This means that you have roughly a bit less than a minute per point.

There are 6 sections with material from the first 7 lectures (sets/graphs/tree, state-spaces, uninformed, informed and adversarial search, and logical agents and knowledge graphs). There are some simple reproduction questions (2-3 points), understanding questions (3 to 4 Points) and do-questions (4 to 8 Points).

Make sure you do not lose too much time on individual questions.

Multiple *Choice* questions have only **one** correct solution. In this case, the options are given by circles. **Multiple *Answers*** questions have one or more correct solution. In this case, the options are given by squares.

The score of Multiple Answer questions is calculated automatically by the system as follows: Score = proportion good * (1 - proportion wrong * (1/a + (((n-k)/n)*(1-1/a))), where n is the total number of alternatives, k the number of correct answers and a a weighting factor set to 2. The more correct answers you give the more points you get, but wrong answers also count negatively.

Set, Graphs and Trees

The following set of questions are about sets, graphs, and trees.

There are 3 questions for a total of 9 points.

Question 1 – Multiple choice – Question-ID: 395667 (2 points)

If U = {1, 2, ... 15}, |A| = 7 and |B| = 7. What is the cardinality of the complement of A, so formally IA'I ?

A 15

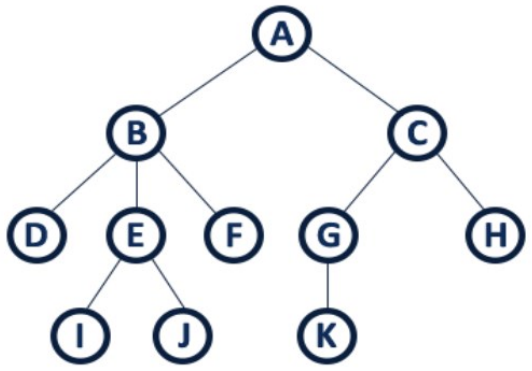
B 8

C 14

D 7

Question 2 – Multiple response – Question-ID: 284679 (3 points)

Which statements of the about the following tree are correct?



A K is the root of the tree

B J is a leaf node

C Node I has depth 2

D A is the root of the tree

E Node I has a depth 3

F E is a leaf node

G The line between G and K is called an edge

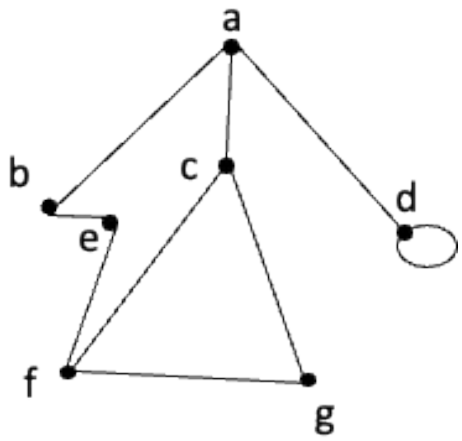
H The line between G and K is called a translation

Question 3 – Multiple choice – Question-ID: 395687 (4 points)

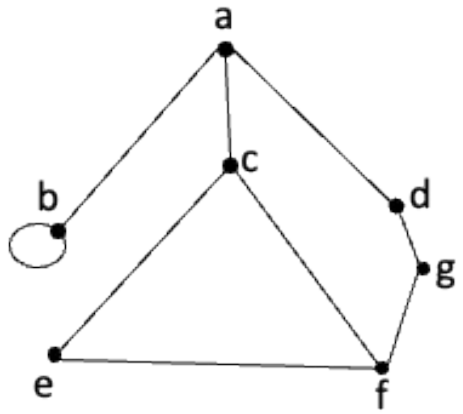
Which graph does the following adjacency matrix represent?

	a	b	c	d	e	f	g
a	0	1	1	0	0	0	1
b	1	0	0	0	0	0	0
c	1	0	0	1	1	0	0
d	0	0	1	1	1	0	0
e	0	0	1	1	0	1	0
f	0	0	0	0	1	0	1
g	1	0	0	0	0	1	0

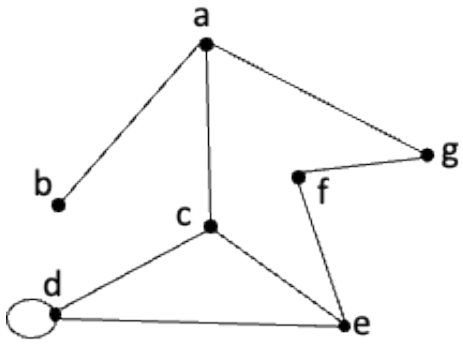
A



B



C



Questions on Uninformed Search

The following set of questions are about uninformed search.

There are 7 questions for a total of 22 points.

Question 4 – Multiple choice – Question-ID: 202997 (2 points)

Which of the following statements is correct?

A There can be many, but only finitely many goal states for each state-space problem.

B There is a unique goal state for each state-space problem

C There might be infinitely many goal states

Question 5 – Multiple choice – Question-ID: 284564 (2 points)

What does fringe do?

A It is a data structure that stores all the states (nodes) that have been visited so far.

B It is a data structure that stores the heuristic values of all the available states (nodes) from the current state.

C It is a pointer to all the nodes in a tree and stores the addresses of all the nodes.

D It is a data structure that stores all available states (nodes) to move to from the current state.

Question 6 – Multiple choice – Question-ID: 395699 (2 points)

Which of the following is a property of depth-first search (DFS)?

A It always finds a solution faster than breadth-first search.

B it always uses an admissible heuristics

C It may not find a goal state

D It is not affected by cycles

E It always guarantees optimality

Question 7 – Multiple choice – Question-ID: 202568 (2 points)

It is better to use backward searching (goal-driven search) when we have many goal states.

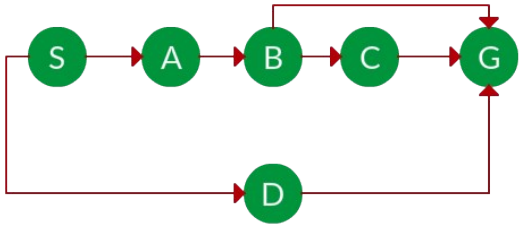
A false

B true

Question 8 – Multiple choice – Question-ID: 284581 (4 points)

To reach goal state G from the initial state S which order of node-exploration deapth-first search (DFS) would follow for the below State-Space?

Nodes are added to the fringe in alphabetical order (i.e. two children X and Y are added in the order X,Y to the fringe).



A S->A->D->B->G

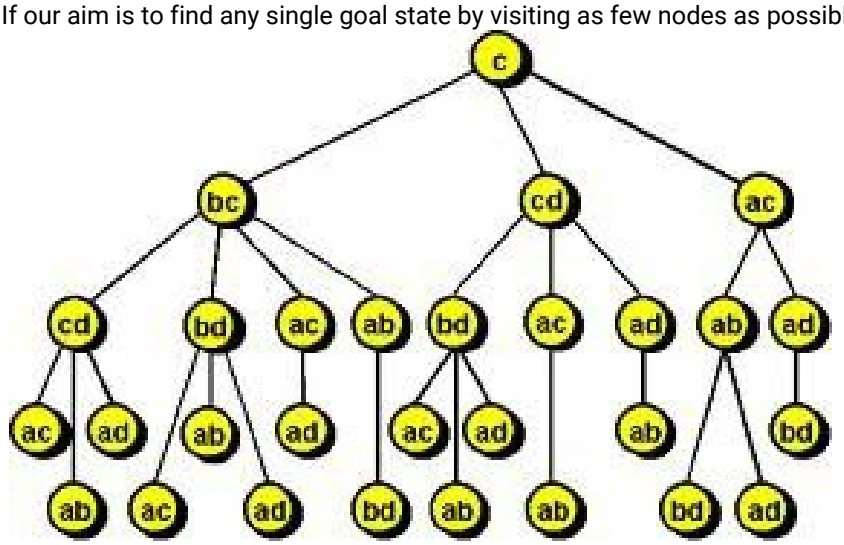
B S -> A -> B -> C -> G

C S -> D -> G

D S ->A-> D -> G

Question 9 – Multiple response – Question-ID: 203001 (4 points)

Let the states labelled with "**bd**" be the goal states, and given the state-space depicted.
If our aim is to find any single goal state by visiting as few nodes as possible, which strategy should we use?



In both cases, the tree is expanded from the left-hand side nodes first.

A breadth first

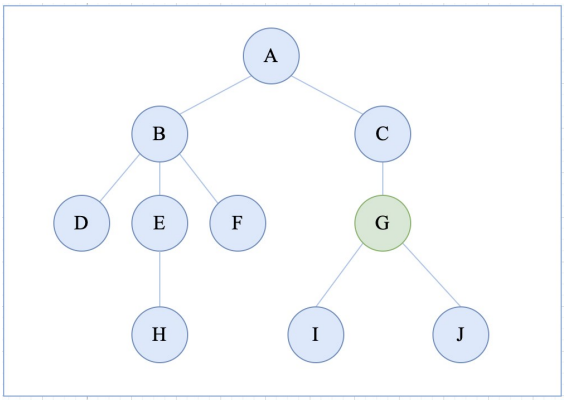
B Goal driven

C Data driven

D depth first

Question 10 – Fill in (multiple) – Question-ID: 343940 (6 points)

Use the following tree with initial state A and goal state G to answer the question about uninformed search. Assume that the tree-search method you're applying adds nodes to the fringe in **alphabetical** order. In other words, the children B,C of A are added to the fringe as B,C.



When applying Breadth-First Search to the tree, which node is removed from the fringe next after F?

Which is the last node to be removed when applying Breadth-First Search?

When applying Depth-First Search, which node is evaluated next after C?

Questions on Informed Search

The following set of questions are about informed search.

There are 7 questions for a total of 27 points.

Question 11 – Multiple choice – Question-ID: 284827 (2 points)

The term Manhatten distance comes from the way you move in New York where you cannot get via the straight-line from one address to the next, but where you have to walk one block straight on, and then a block straight left or right, etc.

Formally, it is the sum of the horizontal and vertical distances between points on a grid. Why is the manhatten distance admissible?

A because the predicted length of getting from one place to the other is the longest possible distance.

B because the length of the search is restricted through the length of the path between two nodes.

C because the predicted length of getting from one place to another is an underestimate of the real distance

D because the actual length of getting from the root to the goal is the shortest possible distance.

Question 12 – Multiple choice – Question-ID: 202977 (2 points)

Which of the following statements is true?

A An algorithm is greedy if it finds the shortest path to a solution.

B Every A* algorithm is a Best-First Search algorithm

C An algorithm is called Best-First search, when it is an A algorithm with admissible heuristics

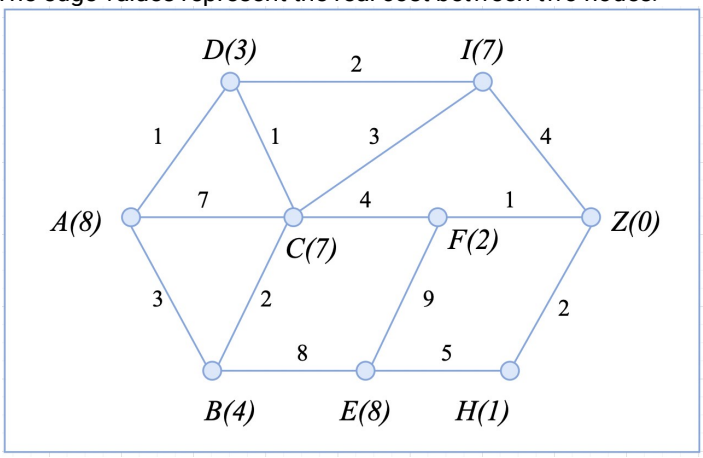
D A hill-climbing algorithm is an A algorithm with admissible heuristics

E Every algorithm that finds an optimal solution is called an A* algorithm

F Every Best-First search algorithm is an A algorithm

Question 13 – Fill in (multiple) – Question-ID: 395057 (4 points)

Consider the following graph. The value in parentheses represents a heuristic function that estimates the distance to the goal. The edge values represent the real cost between two nodes.



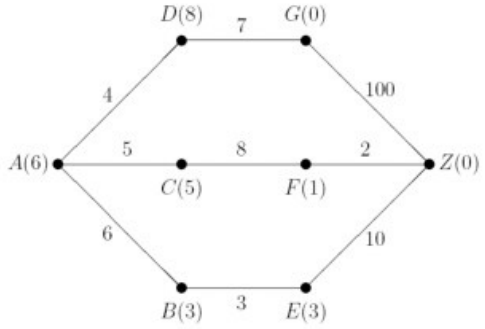
Describe the order in which a Hill-Climbing search algorithm will explore the nodes to reach node Z from node A.
If two nodes have the same heuristic value, choose alphabetic ordering.

Format your answer by separating nodes with comma's and NO spaces, i.e. X,Y,Z

Give up to 8 nodes in the order they are explored (there could be fewer):

Question 14 – Multiple choice – Question-ID: 396540 (5 points)

Consider the following search-space.



Where A is the start, Z the goal position. In brackets is the estimated distance from the goal, on the edges is the real distance travelled. If there is twice the same value, take the first node according to the alphabetic ordering.

What is the order in which the nodes are expanded when greedy Best-First search is used?

A Z,F,C,A

B A,B,A,C,F,Z

C A,B,C,E,Z

D A,B,E,Z

E A,B,C,F,Z

F A,C,F,Z

Question 15 – Multiple response – Question-ID: 284826 (3 points)

Which of the following statements are correct?

A Most informed search algorithms determine the evaluation function by some form of the heuristic.

B Heuristics predict a value for the estimated distance from a goal for each state in the space

C Heuristic are about physical distances.

D In A* search, the time complexity is exponential with the path length

E Heuristic can be wrong

F A hill-climbing algorithm is an A algorithm with admissible heuristics

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Question 16 – Multiple choice – Question-ID: 202981 (3 points)

If we have uniform costs per action and set the heuristic value h(n)=0 for every node n, then A* becomes equal to ...

A hill climbing

B depth-first search

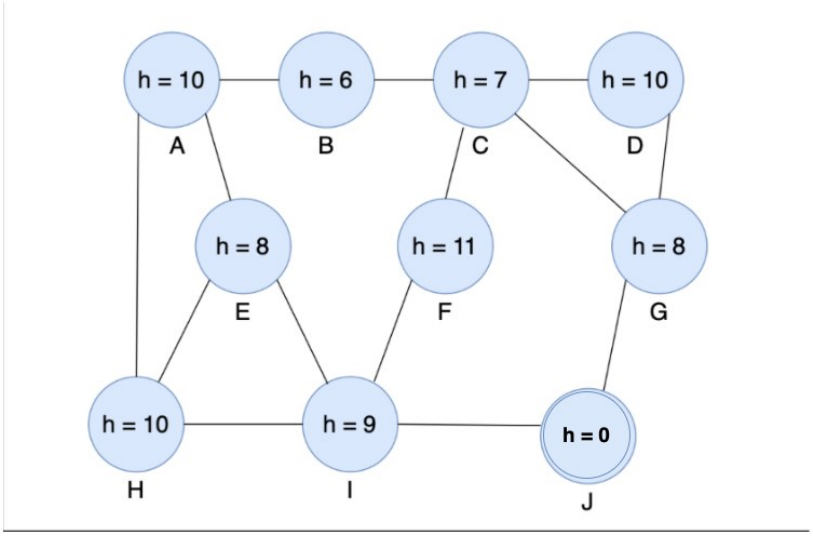
C breadth-first search

D A

Question 17 – Fill in (multiple) – Question-ID: 395056 (8 points)

In which order will A or A* explore the nodes to find a path to the goal state J from the initial node A, when the costs per transition us unique with value 1, and the h values per node denote the estimated distance from the goal state.

If the f values for two nodes are the same, please follow alphabetical order.



Format your answer by separating nodes with comma's and NO spaces, i.e. X,Y,Z

The first 4 nodes visited are [ ] in the order given.
The complete list of nodes visited is: [ ]

Questions on Adversarial Search

The following set of questions are about adversarial search.

There are 4 questions for a total of 20 points.

Question 18 – Multiple choice – Question-ID: 202586 (2 points)

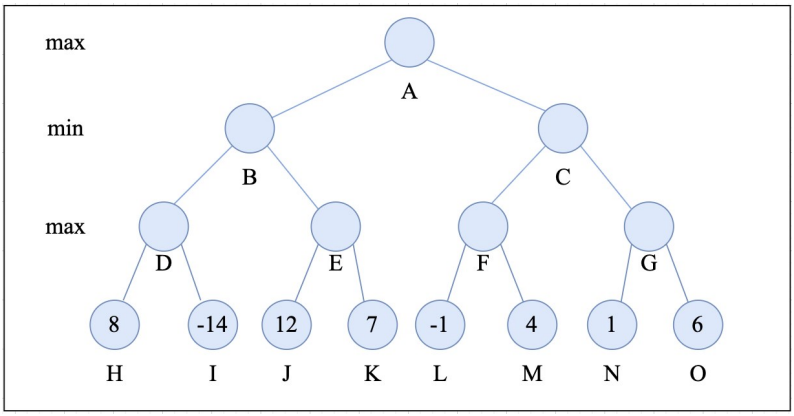
In minimax search with alpha-beta cuts, the alpha value can never [x1] , the beta value can never [x2] .

A [x1] = decrease, [x2] = increase

B [x1] = increase, [x2] = decrease

Question 19 – Fill in (numerical) – Question-ID: 395054 (4 points)

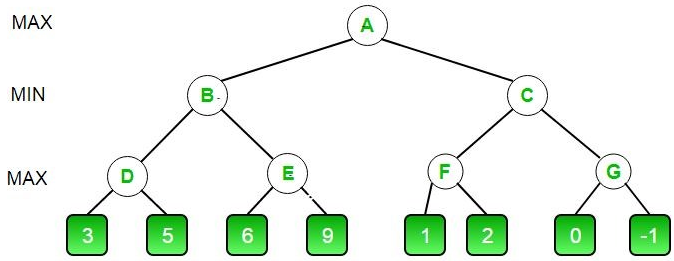
Assume that MAX makes the first move. Given the following tree, what is the value Minimax calculates for node A?



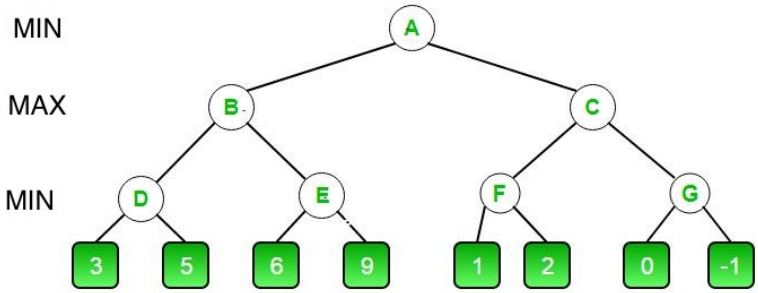
Question 20 – Multiple response – Question-ID: 396565 (6 points)

Consider in the below game trees (**1.** and **2.**). The first move starts from node A.
The maximizer is making the first move in grame tree **1,** while minimizer is making the first move in game tree **2**.
Which of the followinst statements are true?

1.



2.



A In game 2, the value for A is 1

B In game 1, the value for A is 5

C In game 1, the value for A is 9

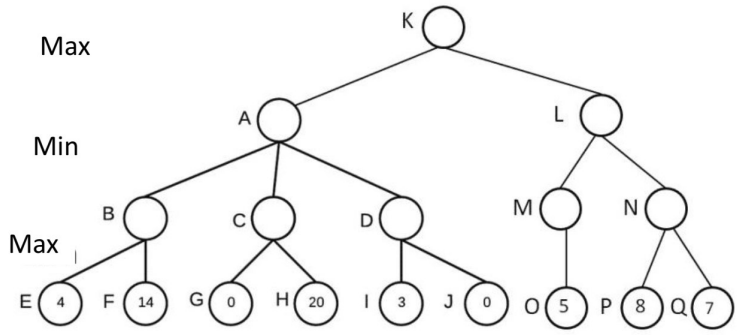
D In game 1, the value for A is 1

E In game 2, the value for A is 3

F In game 2, the value for A is 0

Question 21 – Multiple response – Question-ID: 396613 (8 points)

Consider the following Adversarial Search Space, where a Max player starts the game. .



If you apply MinMax algorithm with Alpha Beta Pruning, which nodes in he tree can you ignore (prune)? More than one answer can be correct.

A L, M, N, O, P, Q

B Q

C M, O

D N, P, Q

E H, I

F D, I, J

G H, J

Questions on Logical Agents

The following set of questions are about logical agents.

There are 9 questions for a total of 32 points.

Question 22 – Match (even) – Question-ID: 284853 (2 points)

Select the correct match of the words with correct statements. (Move the terms on the right over the options on the left)

A Algorithmically determining logical values of sentences and knowledge bases

B Determine the meaning of sentences using interpretations

C Check legal expressions

1 Syntax

2 Calculus

3 Semantics

Question 23 – Fill in (multiple) – Question-ID: 284855 (3 points)

For the statement: $(A \wedge \sim B) \rightarrow C$, complete the below truth table.

Note, ~ stands for negation.

Use the letters T for True and F for False in capital letters!

A	B	~B	$A \wedge \sim B$	C	$(A \wedge \sim B) \rightarrow C$
T	T	F		T	T
T	F	T	T	T	
F	T	F	F		T
F	F	T			

Question 24 – Multiple choice – Question-ID: 343923 (3 points)

Consider the following propositional formula: $p \rightarrow (q \vee \neg r)$.
Which of the following interpretations is NOT a model?

A v(p)=False, v(q)=False, v(r)=False

B v(p)=True, v(q)=False, v(r)=False

C v(p)=True, v(q)=True, v(r)=False

D v(p)=True, v(q)=False, v(r)=True

Question 25 – Multiple choice – Question-ID: 202644 (2 points)

A formula is in clause normal form when it is a [x] of [y] of literals.

A [x] = conjunction, [y] = disjunctions

B [x] = disjunction, [y] = conjunctions

Question 26 – Multiple response – Question-ID: 202875 (4 points)

Consider the following formulas:
-p -> q
q -> - (r v -s)
-q -> -(r v s).

The Clause Normal Form for these formulas contains some of the following clauses. Which ones (please select multiple answers)?

A p v -s

B -q v -r

C -q v s

D p v -r

E p v q

F p v s v r

G -q v r v s

H -p v q

Question 27 – Multiple choice – Question-ID: 396601 (2 points)

Given a knowledge-base in clause normal form {{A,B},{A,-C},{B,-C,-E,D},{C,-D,E}}

Which of the following statements is true for the application of DPLL?

A According to the MOMS heuristics C would be selected first for a value assignment

B According to the MOMS heuristics A would be selected first for a value assignment

C According to the MOMS heuristics B would be selected first for a value assignment

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Question 28 – Multiple choice – Question-ID: 396604 (3 points)

Given a knowledge-base in clause normal form {{A,B},{-B,-C},{B,-C,-E,D},{C,-D,E}}

Which of the following nodes would assigned a value first in a DPLL with MOMs heuristics?

A D

B A

C B

Question 29 – Fill in (multiple) – Question-ID: 284886 (5 points)

Complete the DPLL algorithm steps in the empty place using the option provided below, where ~ denotes negation.

In order to prove satisfiability of a formula F

```
dpll(F, literal){
  remove clauses containing literal
  shorten clauses containing ~literal
  if (F contains no clauses) return [      ];
 if (F contains empty clause)
      return [      ];
  if (F contains a unit or pure L)
      return [      ];
  choose P in F;
  if (dpll(F, ~P)) return true;
  else return [      ];
}
```

Question 30 – Fill in (multiple) – Question-ID: 202972 (8 points)

Given a clause set
{ A v B,
-A v B v C,
-A v -B v C,
-A v B v -C,
A v -B}

Fill in the gaps to prove satisfiability or unsatisfiability with DPLL. If you have a choice, apply an alphabetical order (starting with A), and assing False before True to the variables (and no heuristics). Please keep the clauses in the order given above and do not leave spaces before, between or after propositions.

Assigning the value False to variable A results in the clause set:
[   ]
[   ]
which is a contradiction. So we need to backtrack and assign the value [   ] to [ ].
This results in the knowledge base:
[   ]
[   ]
[   ]

According to the rule described above, assign the value [   ] to variable [ ], which results in the clause set:
[ ]
[ ]
which is again a contradition. So, after this we backtrack on [ ] and give it the value [   ], which results in a single clause
[   ]
This means that the knowledge base is [   ] (satisfiable or unsatisfiable).

In case the knowledge base is satisfiable, provide a model in the following line, otherwise enter XXX (no spaces) in each of the fields.

A=[   ], B=[   ] and C=[   ].

Knowledge Graph and Rulebased Reasoning

There are 2 questions with 10 points

Question 31 – Match (even) – Question-ID: 355320 (4 points)

Using Forward Checking rule-based reasoning, the following rewriting steps are taken to prove whether KB = {A → ¬(¬C ∨ B), A} entails B.
Match each step on the left to the rule on the right that was applied *in order to get to that step.*

For the conclusion statement, match the correct entailment statement to the "Correct conclusion?" box and the incorrect entailment statement to "Incorrect conclusion?"

A ¬(¬C ∨ B)

B (C ∧ ¬B)

C A, ¬B

D Correct conclusion:

E Incorrect conclusion:

1 A is *not* entailed by the knowledge base.

2 A is entailed by the knowledge base.

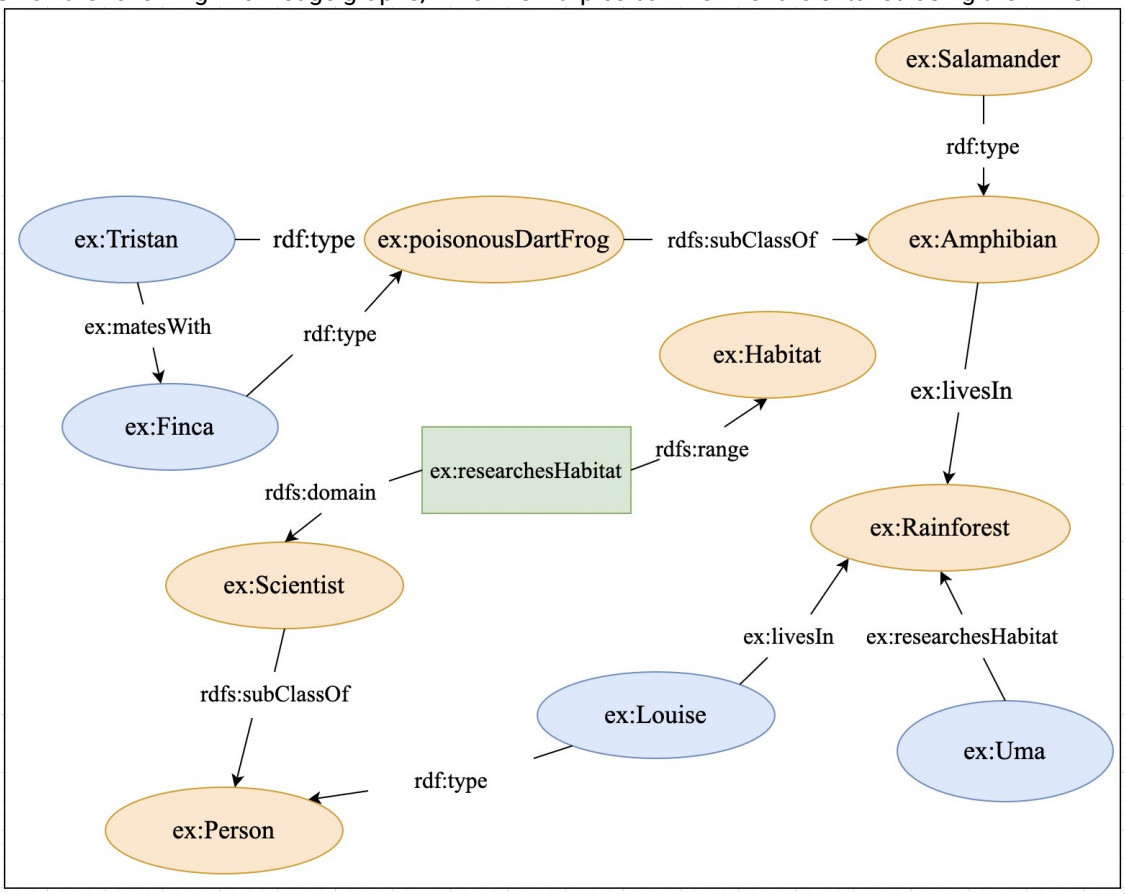
3 Modus Ponens

4 de Morgan

5 and-Elimination

Question 32 – Multiple response – Question-ID: 355316 (6 points)

Given the following knowledge graphs, which new triples can we infer are entailed using the RDFS inferencing rules?



A ex:Tristan rdf:subClassOf ex:Amphibian .

B ex:Tristan rdf:type ex:Amphibian .

C ex:Uma rdf:type ex:Person .

D ex:Rainforest rdf:type ex:Habitat .

E ex:Finca ex:livesIn ex:Rainforest

F ex:Louise rdf:type ex:Scientist

G ex:Scientist ex:researchesHabitat ex:poinsonousDartFrog

H ex:Salamander ex:livesIn ex:Rainforest

End of test Intelligent Systems COMPUTERROOMS